COMPOSITION OF DERIVED TENSOR FUNCTORS THROUGH AN INTERMEDIATE DIFFERENTIAL GRADED ALGEBRA IS A DERIVED TENSOR WITH A BIMODULE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-dga/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the K-flat replacement with its equivalence and restriction quasi-inverse, the three-composition reduction, the dispatch of the two compositions, the property-(P) resolution P = L tensor_R B with the rewriting of the comparison map, and the final reduction through more-algebra-lemma-K-flat-quasi-isomorphism to the standing hypothesis are all retained verbatim.

Editorial note. The label lemma-compose-tensor-functors-tor names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- K-flat replacement of the intermediate algebra. Same chapter, Lemmas lemma-K-flat-resolution and lemma-qis-equivalence. Lets the intermediate algebra (B, d) be replaced by a K-flat quasi-isomorphic (B', d) whose derived category is equivalent to D(B, d) with restriction as quasi-inverse.
- Reduction lemmas for plain versus derived tensor. Same chapter, Lemmas lemma-compose-tensor-functors, lemma-compose-tensor-functors-general-algebra; More on Algebra, Lemma more-algebra-lemma-K-flat-quasi-isomorphism. Handle the two reduced compositions, identify the target map for the K-flat case, and finally replace a quasi-isomorphism tensored with a K-flat complex.

Source reference. Original equation 0BZ4: sources/stacks-dga/pinned-original-source.tex, source lines 5941–5941. Prerequisite pointer invoked at proof line(s) 7255; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 06Y4: sources/stacks-dga/pinned-original-source.tex, source lines 7157–7157. Prerequisite pointer invoked at proof line(s) 7230; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09S4: sources/stacks-dga/pinned-original-source.tex, source lines 6067–6067. Prerequisite pointer invoked at proof line(s) 7245; it names

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a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0BZ3: sources/stacks-dga/pinned-original-source.tex, source lines 5867–5867. Prerequisite pointer invoked at proof line(s) 7249; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0BZ5: sources/stacks-dga/pinned-original-source.tex, source lines 5959–5959. Prerequisite pointer invoked at proof line(s) 7257; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09S6: sources/stacks-dga/pinned-original-source.tex, source lines 6725–6725. Prerequisite pointer invoked at proof line(s) 7233; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 06Y0: sources/stacks-chow/tags.txt, source lines 8869–8869. Prerequisite pointer invoked at proof line(s) 7274; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let R be a ring. Let (A, d), (B, d), and (C, d) be differential graded R-algebras. Assume $A \otimes_R C$ represents $A \otimes^{\mathbf{L}}_R C$ in $D(R)$. Let N be a differential graded (A, B)-bimodule. Let N' be a differential graded (B, C)-bimodule. Then the composition*

$$D(A, d) \xrightarrow{-\otimes^{\mathbf{L}}_A N} D(B, d) \xrightarrow{-\otimes^{\mathbf{L}}_B N'} D(C, d)$$

is isomorphic to $- \otimes^{\mathbf{L}}_A N''$ for some differential graded (A, C)-bimodule N''.

Proof. Using Lemma 06Y4 we choose a quasi-isomorphism $(B', \mathrm{d}) \to (B, \mathrm{d})$ with B' K-flat as a complex of R-modules. By Lemma 09S6 the functor $- \otimes^{\mathbf{L}}_{B'} B : D(B', \mathrm{d}) \to D(B, \mathrm{d})$ is an equivalence with quasi-inverse given by restriction. Note that restriction is canonically isomorphic to the functor $- \otimes^{\mathbf{L}}_B B : D(B, \mathrm{d}) \to D(B', \mathrm{d})$ where B is viewed as a (B, B')-bimodule. Thus it suffices to prove the lemma for the compositions

$$D(A) \to D(B) \to D(B'), \quad D(B') \to D(B) \to D(C), \quad D(A) \to D(B') \to D(C).$$

The first one is Lemma 09S4 because B' is K-flat as a complex of R-modules. The second one is true because $B \otimes^{\mathbf{L}}_B N' = N' = B \otimes_B N'$ and hence Lemma 0BZ3 applies. Thus we reduce to the case where B is K-flat as a complex of R-modules.

Assume B is K-flat as a complex of R-modules. It suffices to show that (0BZ4) is an isomorphism, see Lemma 0BZ5. Choose a quasi-isomorphism $L \to A$ where L is a differential graded R-module which has property (P). Then it is clear that $P = L \otimes_R B$ has property (P) as a differential graded B-module. Hence we have to show that $P \to A \otimes_R B$ induces a quasi-isomorphism

$$P \otimes_B (B \otimes_R C) \longrightarrow (A \otimes_R B) \otimes_B (B \otimes_R C)$$

We can rewrite this as

$$P \otimes_R B \otimes_R C \longrightarrow A \otimes_R B \otimes_R C$$

Since B is K-flat as a complex of R-modules, it follows from More on Algebra, Lemma 06Y0 that it is enough to show that

$$P \otimes_R C \to A \otimes_R C$$

is a quasi-isomorphism, which is exactly our assumption. □